POLITECNICO MILANO 1863

SCHOOL OF INDUSTRIAL AND INFORMATION ENGINEERING

MASTER OF SCIENCE IN ELECTRICAL ENGINEERING

Electric Power Systems Project 2023-2024: Small Signal Stability Analysis

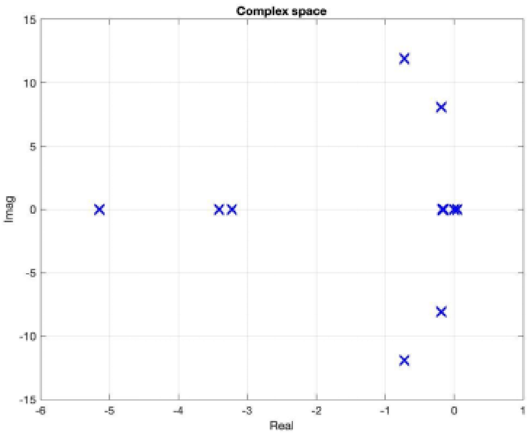
Academic Year 2023 - 2024

STEP 1

A) By Plotting the eigenvalues in the complex space with all the exciters and all the PSS switched off:

	1	2	3	4	5	6	7	8	9	10	11	12
1	0	0	0	314.1593	0	0	0	0	0	0	0	0
2	0	0	0	0	314.1593	0	0	0	0	0	0	0
3	0	0	0	0	0	314.1593	0	0	0	0	0	0
4	-0.0505	0.0276	0.0229	0	0	0	0.0473	-0.0056	-0.0077	-0.0042	0.0326	0.0251
5	0.1417	-0.2450	0.1033	0	0	0	-0.1327	0.0725	-0.0629	0.0036	-0.2604	0.0789
6	0.2016	0.1796	-0.3812	0	0	0	-0.1880	-0.0699	0.1803	0.0383	0.1695	-0.3537
7	0	0	0	0	0	0	-3.2258	0	0	0	0	0
8	-0.5611	1.0956	-0.5345	0	0	0	0.5638	-4.5627	0.9824	1.6763	0.4533	0.3865
9	-0.8847	-1.1099	1.9946	0	0	0	0.8671	1.3189	-4.9972	1.6809	0.0876	0.3777
10	0.0012	-0.0017	4.7571e-...	0	0	0	-0.0075	0.0139	0.0107	-0.1260	0.0154	0.0136
11	0.2003	-0.3297	0.1294	0	0	0	-0.1825	0.0238	0.0083	0.2291	-0.4375	0.2041
12	0.2237	0.1549	-0.3786	0	0	0	-0.2028	0.0625	0.0106	0.2982	0.3030	-0.5463

Table 1: Dynamic matrix A



Eigenvalues					
No.	Real	Imaginary	Frequency (Hz)	Damping Ratio	
1,2	-0.724	±11.9282	1.8984	0.061	Mode 1
3,4	-0.1914	±8.0860	1.2869	0.024	Mode 2
5	-5.1453	-	-	1	
6	-3.411	-	-	1	
7	-0.1554	-	-	1	
8	-0.1739	-	-	1	
9	0.0468	-	-	1	
10,11	0	±0.0018	-0.0003	-1	
12	-3.2258	-	-	0	

Graph 1: complex space

Table 2: eigenvalues

*Zero on number 11 is an Error by MATLAB

1.1 A): How many electromechanical modes have you obtained? Provide the damping ratio and the frequency of the electromechanical modes identified.

To identify electromechanical modes:

- **Damping and frequency**: Electromechanical modes typically have low damping ratios (less than 10%) and low frequencies (less than 5 Hz) compared to other modes in the system.
- **Participation factors**: Electromechanical modes usually have high participation factors from the generator rotor angles and speeds, which indicate the contribution of the state variables to the mode shape.

The presence of a pair of conjugate eigenvalues with a non-zero imaginary component indicates the existence of an oscillation mode. By examining the eigenvalues presented in the aforementioned table, it is observed that there are two oscillation modes, which align with the presence of N-1 oscillation modes if there are N generators. These modes are characterized by the following values: D=0.0610, f=1.8984 and D=0.0240, f=1.2869. However, it should be noted that for the case where f=-0.003, the imaginary part of the eigenvalue can be considered zero due to an error in MATLAB.

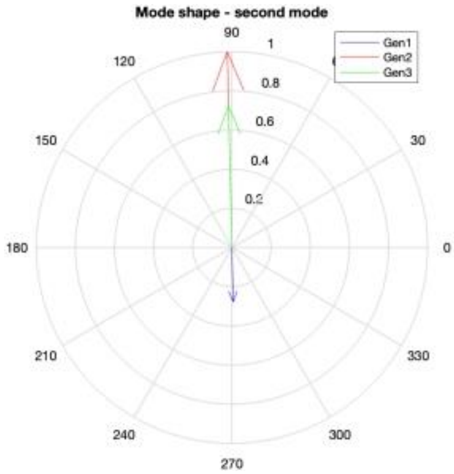
eigenvalues : $\Omega = \alpha \pm j\beta$ Frequencies: $f = \frac{\beta}{2\pi}$ Damping: D=($\frac{-\alpha}{\sqrt{\alpha^2+\beta^2}}$)

The calculated values of damping and frequency is as follow:

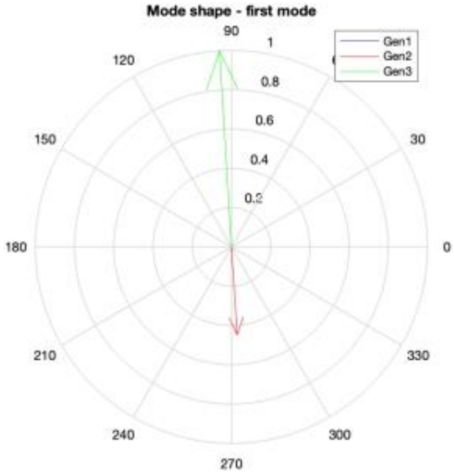
```
frequency =        damping =

    1.8984            0.0610
    1.2869            0.0240
         0            1.0000
         0            1.0000
         0           -1.0000
   -0.0003                 0
```

1.2 A): Which generator oscillates against which? Can you classify the modes? Analyze the results by plotting the mode shapes:

By drawing the mode shape of the generator rotor angular velocity, the following graphs are obtained.



Graph 2: Mode shape-second mode



Graph 3: Mode shape-first mode

The above graphs show the oscillation patterns of Gen 1, Gen 2, and Gen 3 in two different modes. In the first mode, Gen 2 and Gen 3 oscillate almost out of phase with each other, while Gen 1 has a smaller amplitude (Local). In the second mode, Gen 1 oscillates in the opposite phase of Gen 2 and Gen 3, and Gen 2 has the largest amplitude among the three (Inter_area).

Mode	Dominant States	Oscillation Phase	
1	Δw and Δδ of G2, G3	G2	G3
2	Δw and Δδ of G1, G2, G3	G2, G3	G1
		Anti_Phase	

Table 3

1.3 A) Can you classify the remaining non-electromechanical, oscillatory modes?

Pkh	Pk1		Pk2									
G1	0.00553	0.00553	0.39227	0.39227	3.72E-06	0.001687	0.00059	5.41E-06	0.03817	1	1	2.39E-18
G2	0.3512	0.3512	1	1	0.01593	0.002332	0.00032	0.00029	0.02772	0.22082	0.22082	6.42E-18
G3	1	1	0.29867	0.29867	0.02417	0.011326	1.30E-05	0.00079	0.01022	0.12412	0.12412	5.38E-17
G1	0.00553	0.00553	0.39227	0.39227	3.72E-06	0.001687	0.00059	5.41E-06	0.03817	1	1	7.02E-15
G2	0.3512	0.3512	1	1	0.01593	0.002332	0.00032	0.00029	0.02772	0.22082	0.22082	4.77E-15
G3	1	1	0.29867	0.29867	0.02417	0.011326	1.30E-05	0.00079	0.01022	0.12412	0.12412	1.96E-15
G1	0	0	0	0	0	0	0	0	0	0	0	1
G2	0.01896	0.01896	0.02161	0.02161	1	0.886699	0.00417	0.00468	0.00063	1.82E-07	1.82E-07	6.97E-17
G3	0.09177	0.09177	0.00709	0.00709	0.90669	1	0.00017	0.02806	0.00048	1.94E-07	1.94E-07	1.33E-16
G1	1.30E-05	1.30E-05	2.49E-05	2.49E-05	6.32E-06	0.007511	1	0.05327	0.40393	0.00014	0.00014	5.73E-16
G2	0.01833	0.01833	0.04059	0.04059	0.00808	0.001543	0.33341	0.29903	1	0.00022	0.00022	7.77E-17
G3	0.03533	0.03533	0.01074	0.01074	0.02032	0.002598	0.00753	1	0.48361	9.56E-05	9.56E-05	7.39E-16
	Mode 1 (Local Area)		Mode 2 (Inter_area)									

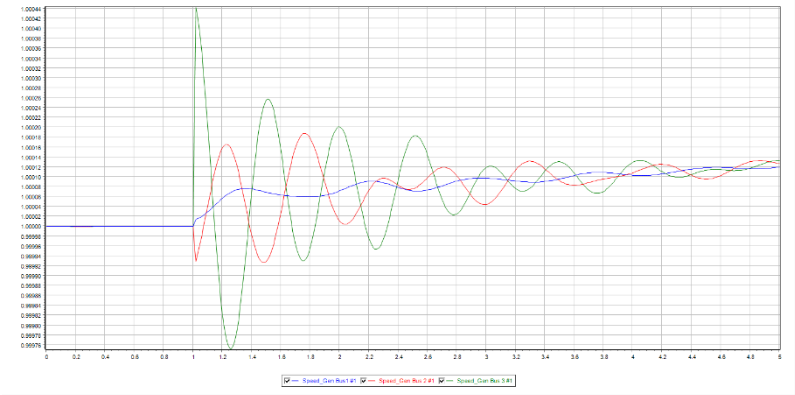
Table 4: Participation factors

The participation factor characterizes the correlation between state variables and oscillation patterns. Through the participation factor, in mode 1, the perturbation only affects Gen2 (0.3512) and Gen3 (1), and Gen3 is the most affected. However, the effect of mode 1 on Gen1 is very small and negligible (0.0055). It is consistent with the fact that the vector modulus value of mode1 representing Gen1 in mode shape analysis is almost 0.

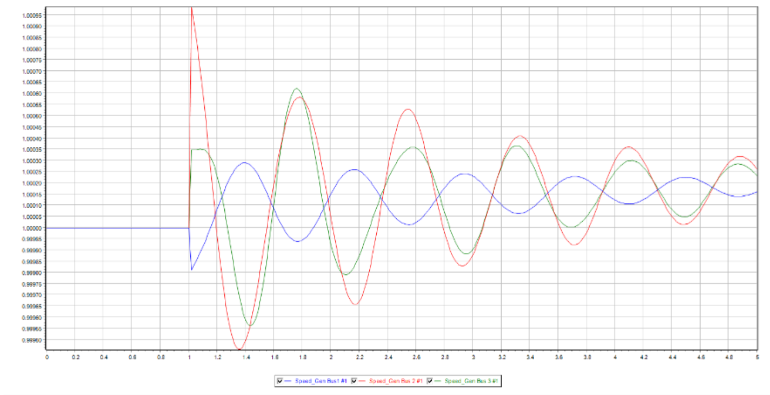
In mode 2, the disturbance influences all three generators, which are 0.3923, 1, and 0.2987, respectively, and mode 2 has the greatest effect on the state variables of Gen2, which is consistent with the mode shape analysis. From the above analysis, mode 2 influences all generator state variables in the system, which is inter-area mode. Mode 1 only affects some of the generator state variables in the system, so it is local mode.

1.B) After a transient (line opening), the speeds of the generators are altered. Plot the time domain behavior in Power World: is it coherent with your modal analysis?

By plotting the time domain in the Power World, the following results are obtained:



Graph 3: Mode shape-first



Graph 4: Mode shape-second

The result obtained from Power World is as same as the result that obtained from the MATLAB.

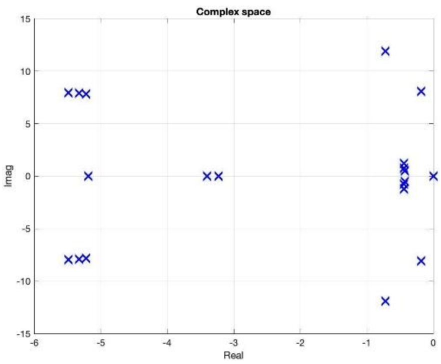
Object Pretty	Time (Cycles)	Time (Seconds)	Object	Description
Line Bus 8 To Bus 9 CKT 1	50	1	Branch '8' '9' '1'	OPEN BOTH
Line Bus 8 To Bus 9 CKT 1	51	1.02	Branch '8' '9' '1'	CLOSE BOTH
Object Pretty	Time (Cycles)	Time (Seconds)	Object	Description
Line Bus 5 To Bus 7 CKT 1	50	1	Branch '5' '7' '1'	OPEN BOTH
Line Bus 5 To Bus 7 CKT 2	51	1.02	Branch '5' '7' '1'	CLOSE BOTH

Table 5

Step 2

2.1 A) Do you get any improvement in the damping ratios of the electromechanical modes? Comment the results.

By Switching on all the exciters and redo the analysis of step one the outcome is:



Eigenvalues			Frequency (Hz)	Damping Ratio	
No.	Real	Imaginary			
1,2	-0.7178	±11.9241	1.8978	0.06	Mode 1 (electromechanical)
3,4	-0.1833	±8.0757	1.2853	0.023	Mode 2 (electromechanical)
5,6	-5.4872	±7.9488	1.2651	0.568	Mode 3
7,8	-5.3264	±7.9213	1.2607	0.558	Mode 4
9,10	-5.2226	±7.8166	1.2441	0.556	Mode 5
11	-5.1875	_	0	1	
12	-3.4032	_	0	1	
13,14	-0.4443	±1.2106	0.1927	0.345	Mode 6
15,16	-0.4393	±0.7392	0.1177	0.511	Mode 7
17,18	-0.4257	±0.4960	0.0789	0.651	Mode 8
19,20	0	±0.0018	0.0003	0	
21	-3.2258	_	0	1	

Graph 5: complex space after switching on exciters

Table 6: eigenvalues after switching on exciters.

To identify electromechanical modes:

- **Damping and frequency**: Electromechanical modes typically have low damping ratios (less than 10%) and low frequencies (less than 5 Hz) compared to other modes in the system.
- **Participation factors**: Electromechanical modes usually have high participation factors from the generator rotor angles and speeds, which indicate the contribution of the state variables to the mode shape.

As shown in *Table 6*, the first and second modes are electromechanical modes, which are identified by their low damping, high participation factors, and large impact on the generators. The other modes (3-8) are oscillation modes, which have higher damping and lower participation factors.

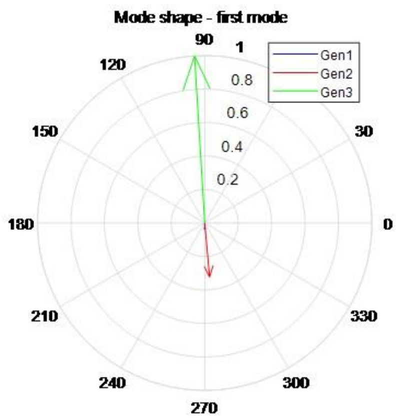
2.2 A) How many electromechanical modes have you obtained? Provide the damping ratio and the frequency of the electromechanical modes identified.

Referring the *Table 4* there are there are 2 electromechanical modes of oscillation due to the higher frequency and lower damping ratio and also there are 6 other oscillations The recalculated values of damping and frequency is as follow:

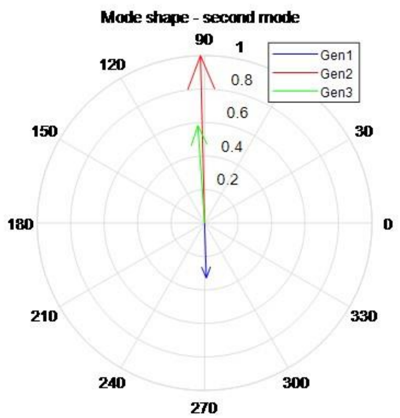
```
frequency =        damping =

    1.8978             0.0600
    1.2853             0.0230
    1.2651             0.5680
    1.2607             0.5580
    1.2441             0.5560
         0             1.0000
    0.1927             0.3450
    0.1177             0.5110
    0.0789             0.6510
    0.0003                  0
         0             1.0000
```

2.3 A) Which generator oscillates against which? Can you classify the modes? Analyze the results by plotting the mode shapes:

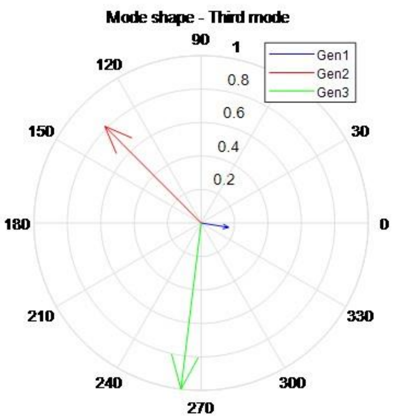
By drawing the mode shapes of the generator rotor angular velocity, the following graphs obtain.



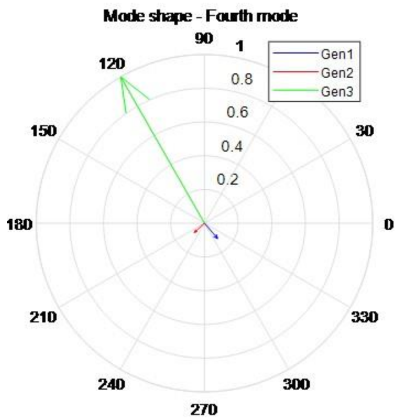
Graph 6: Mode shape-first mode



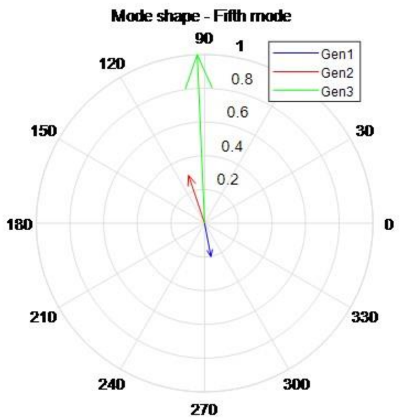
Graph 7: Mode shape-second mode



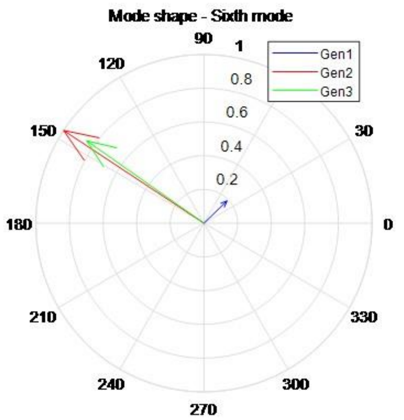
Graph 8: Mode shape-third mode



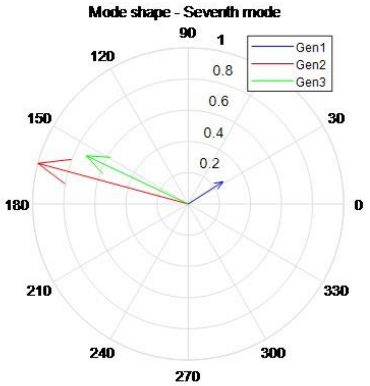
Graph 9: Mode shape-fourth mode



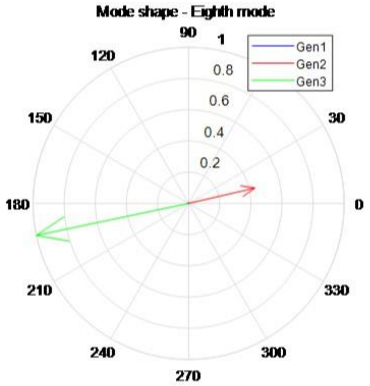
Graph 10: Mode shape-fifth mode



Graph 11: Mode shape-sixth mode



Graph 12: Mode shape-seventh mode



Graph 13: Mode shape-eighth mode

Mode	Dominant States	Oscillation Phase	
1	Δw and Δб of G2, G3	G2	G3
2	Δw and Δб of G1, G2, G3	G2, G3	G1
3	Δw and Δб of G1, G2, G3	_	_
4	Δw and Δб of G1, G2, G3	G3	G1
5	Δw and Δб of G1, G2, G3	G2	G1
6	Δw and Δб of G1, G2, G3	_	_
7	Δw and Δб of G1, G2, G3	_	_
8	Δw and Δб of G2, G3	G3	G2
		Anti_Phase	

Table 7

The oscillation modes of the generators are summarized as follows:

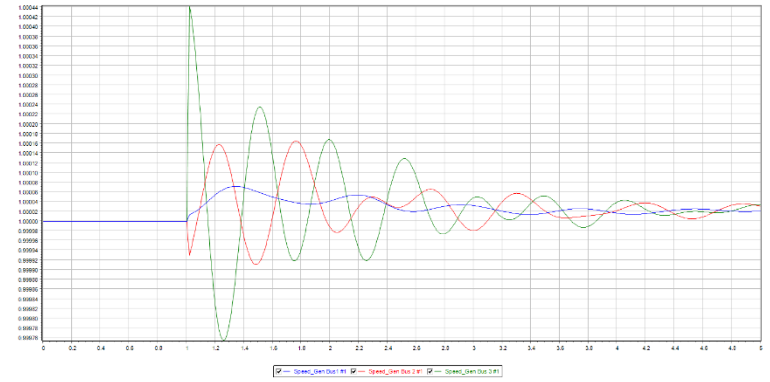
- ***Mode 1****:* A local mode, where G3 oscillates out of phase with G2, and G1 has negligible oscillation.
- ***Mode 2****:* An inter-area mode, where G3 and G2 oscillate in phase with each other, and opposite to G1.
- ***Mode 3****:* A global mode, where all generators oscillate with a phase difference of about 120 degrees.
- ***Mode 4****:* A local mode, where G3 has a large oscillation, and G1 and G2 have very small oscillations.
- ***Mode 5****:* An inter-area mode, similar to mode 2, but with a smaller oscillation amplitude for G3 and G2.
- ***Mode 6****:* A global mode, like mode 3, but with a smaller oscillation amplitude for all generators.
- ***Mode 7****:* A global mode, almost identical to mode 6.
- ***Mode 8****:* An inter-area mode, where G2 and G3 oscillate out of phase with each other, and G1 has negligible oscillation.

Table 8: Participation factors after switching exciters.

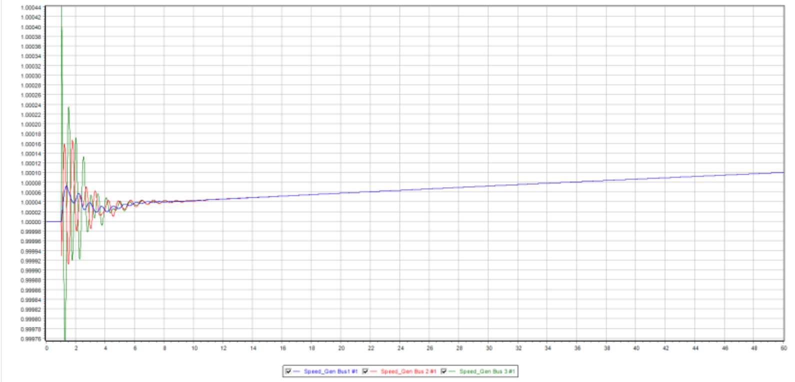
Pkh	Pk1		Pk2		Pk3		Pk4		Pk5				Pk6		Pk7				PK8		
G1	0.00555	0.00555	0.39142	0.39142	0.00039	0.00039	0.00049	0.00049	0.00036	0.00036	3.11E-06	0.00133	0.00082	0.00082	0.00092	0.00092	8.13E-06	8.13E-06	1	1	5.69E-17
G2	0.35093	0.35093	1	1	0.00272	0.00272	0.00012	0.00012	0.00036	0.00036	0.0149	0.002	0.00291	0.00291	0.00301	0.00301	0.00177	0.00177	0.19542	0.19542	1.36E-16
G3	1	1	0.29658	0.29658	0.00091	0.00091	0.00383	0.00383	0.00105	0.00105	0.02366	0.01069	0.00206	0.00206	0.00095	0.00095	0.00432	0.00432	0.11308	0.11308	8.28E-17
G1	0.00555	0.00555	0.39142	0.39142	0.00039	0.00039	0.00049	0.00049	0.00036	0.00036	3.11E-06	0.00133	0.00082	0.00082	0.00092	0.00092	8.13E-06	8.13E-06	1	1	1.60E-14
G2	0.35093	0.35093	1	1	0.00272	0.00272	0.00012	0.00012	0.00036	0.00036	0.0149	0.002	0.00291	0.00291	0.00301	0.00301	0.00177	0.00177	0.19542	0.19542	9.68E-15
G3	1	1	0.29658	0.29658	0.00091	0.00091	0.00383	0.00383	0.00105	0.00105	0.02366	0.01069	0.00206	0.00206	0.00095	0.00095	0.00432	0.00432	0.11308	0.11308	4.05E-15
G1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
G2	0.01856	0.01856	0.01737	0.01737	0.00449	0.00449	0.00167	0.00167	0.00325	0.00325	1	0.90261	0.0464	0.0464	0.03892	0.03892	0.02802	0.02802	3.74E-09	3.74E-09	4.12E-16
G3	0.09045	0.09045	0.00518	0.00518	0.00142	0.00142	0.00777	0.00777	0.00154	0.00154	0.9357	1	0.02597	0.02597	0.02104	0.02104	0.11336	0.11336	3.80E-09	3.80E-09	4.06E-16
G1	3.05E-05	3.05E-05	0.00015	0.00015	0.00057	0.00057	0.003	0.003	0.05076	0.05076	4.76E-06	0.00301	1	1	1	1	0.00821	0.00821	1.58E-07	1.58E-07	1.11E-18
G2	0.01885	0.01885	0.04041	0.04041	0.03508	0.03508	0.00489	0.00489	0.00636	0.00636	0.01593	0.00178	0.67105	0.67105	0.77764	0.77764	0.42548	0.42548	3.40E-06	3.40E-06	5.25E-17
G3	0.03385	0.03385	0.01048	0.01048	0.00195	0.00195	0.03086	0.03086	0.0103	0.0103	0.03012	0.00617	0.3841	0.3841	0.22344	0.22344	1	1	1.83E-06	1.83E-06	6.96E-17
G1	2.35E-05	2.35E-05	0.00012	0.00012	0.01705	0.01705	0.1718	0.1718	0.97327	0.97327	7.88E-08	7.42E-05	0.15519	0.15519	0.1401	0.1401	0.00113	0.00113	1.85E-08	1.85E-08	1.13E-17
G2	0.00079	0.00079	0.00375	0.00375	0.98351	0.98351	0.06761	0.06761	0.05769	0.05769	5.55E-05	5.14E-05	0.10176	0.10176	0.10833	0.10833	0.05759	0.05759	7.12E-08	7.12E-08	8.49E-17
G3	0.00225	0.00225	0.00147	0.00147	0.09649	0.09649	0.97619	0.97619	0.15078	0.15078	6.42E-05	5.50E-05	0.05922	0.05922	0.0323	0.0323	0.14392	0.14392	2.87E-08	2.87E-08	9.95E-17
G1	2.27E-05	2.27E-05	0.00011	0.00011	0.01761	0.01761	0.17684	0.17684	1	1	7.67E-07	1.82E-06	0.10406	0.10406	0.09148	0.09148	0.00073	0.00073	1.24E-08	1.24E-08	8.41E-19
G2	0.00077	0.00077	0.00355	0.00355	1	1	0.06857	0.06857	0.05843	0.05843	0.00037	1.74E-05	0.07994	0.07994	0.08398	0.08398	0.04447	0.04447	5.62E-08	5.62E-08	9.17E-18
G3	0.00218	0.00218	0.00136	0.00136	0.09917	0.09917	1	1	0.1542	0.1542	0.00057	4.53E-06	0.0417	0.0417	0.02225	0.02225	0.09848	0.09848	2.03E-08	2.03E-08	2.72E-18
G1	2.07E-06	2.07E-06	2.99E-05	2.99E-05	0.005	0.005	0.05155	0.05155	0.30565	0.30565	1.27E-05	0.02553	0.73994	0.73994	0.77698	0.77698	0.0065	0.0065	7.41E-08	7.41E-08	2.91E-16
G2	6.96E-05	6.96E-05	0.0009	0.0009	0.28854	0.28854	0.02029	0.02029	0.01812	0.01812	0.00894	0.01767	0.4852	0.4852	0.60078	0.60078	0.3329	0.3329	2.86E-07	2.86E-07	1.21E-15
G3	0.0002	0.0002	0.00035	0.00035	0.02831	0.02831	0.29294	0.29294	0.04735	0.04735	0.01034	0.01891	0.28236	0.28236	0.1791	0.1791	0.83193	0.83193	1.15E-07	1.15E-07	4.83E-16
	Mode 1		Mode 2		Mode 3		Mode 4		Mode 5				Mode 6		Mode 7				Mode 8		

Regarding *Table 8*, it can be understood that it can be seen from the mode shape and participation factor that in mode1, only Gen2 and Gen3 oscillate. Gen1 has no oscillation, so it is a local mode. In mode2, all three generators have oscillations, so it is an inter-area. mode. And it can be seen from the participation factor that the sum is the Dominant States of MODE1 and 2. Therefore, the generator rotor angular velocity is mainly related to these two modes.

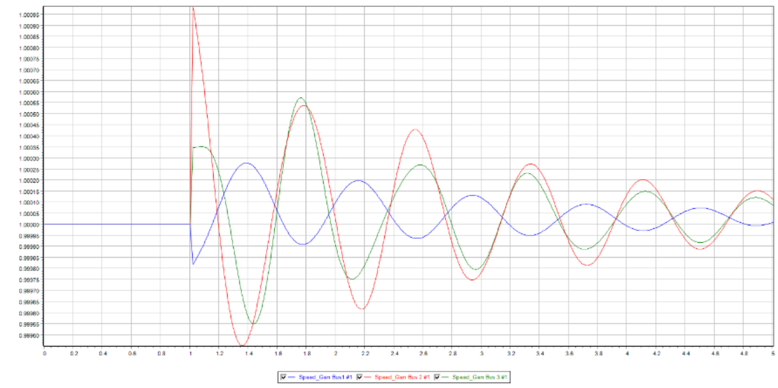
After plotting the time domain in the Power World, the following results are obtained:



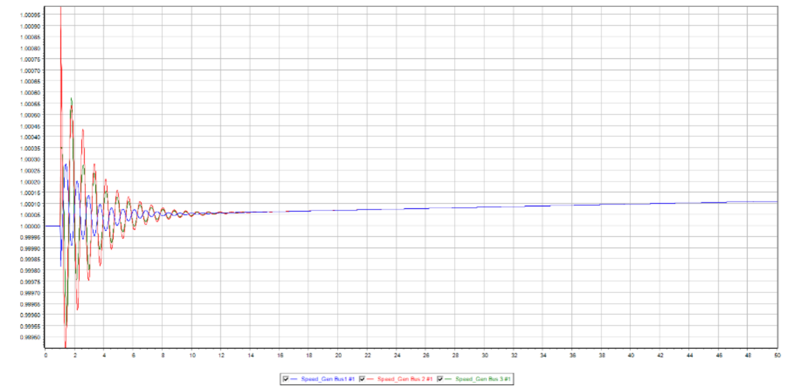
Graph 14: Mode shape-first



Graph 15: Mode shape-first for 50s



Graph 16: Mode shape-second



Graph 17: Mode shape-second for 50s

The plots show that the results for mode 1 and 2 are consistent with MATLAB. However, when the simulation time is extended to 50s, we observe that the negative synchronizing power causes the system to lose synchronism in the long run after adding AVR fault 1, despite the sufficient damping power. This is because high gain excitation systems can reduce the damping torques significantly, leading to instability if the net damping torque is negative, according to the theory. A possible solution to this instability problem is to install Power System Stabilizers, which are continuous controllers that can generate positive damping torques on the generators.

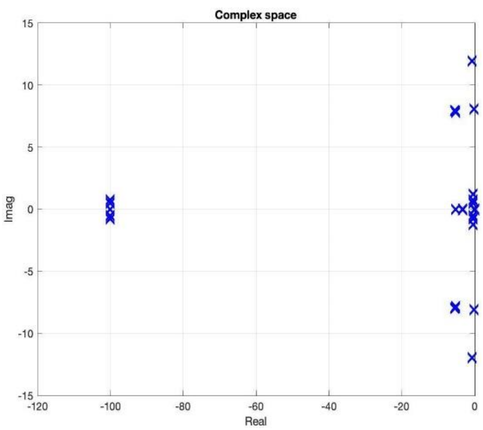
8-9: The system reaches a new equilibrium point, which is unstable.
5-7: The system reaches a new equilibrium point at 1.0001, which is also unstable.

2.4 A) Do you get any improvement in the damping ratios of the electromechanical modes? Comment the results.

Yes. Through the participation factors charts got from two case. it can be seen that the application of AVR gives a negative contribution to damping, but only has little effect on the damping ratios, the damping ratio of mode1 is decreased from 0.61 to 0.60, and the damping ratio of mode2 is decreased from 0.24 to 0.23.

Step 3

3 A) Switch on also all the PSS and redo the analysis of point 1:
A) By Plotting the eigenvalues in the complex space with all the exciters and all the PSS switched on:



No.	Eigenvalues Real	Imaginary	Frequency (Hz)	Damping Ratio	
1,2	-100.0100	±0.77596e-01	0.1235	1	Mode1
3,4	-100.0000	±5.0778e-01	0.0808	1	Mode2
5	-100.0300	_	0	1	
6	-99.9705	_	0	1	
7,8	-0.7299	±11.9506	1.902	0.061	Mode3 (electromechanical)
9,10	-0.1942	±8.0823	1.2863	0.024	Mode4 (electromechanical
11,12	-5.4730	±7.9429	1.2642	0.567	Mode5
13,14	-5.2223	±7.8156	1.2439	0.556	Mode6
15,16	-5.3091	±7.9169	1.26	0.557	Mode7
17	-5.1877	_	0	1	
18	-3.4032	_	0	1	
19,20	-0.4443	±1.2104	0.1926	0.345	Mode8
21,22	-0.4393	±0.7391	0.1176	0.511	Mode9
23,24	-0.4257	±0.4958	0.0789	0.51	Mode10
25,26	0.0000	±0.0018	0.0003	0	
27	-0.0998	_	0	1	
28	-0.1000	_	0	1	
29	-0.1000	_	0	1	

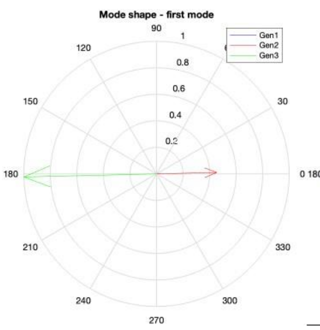
Graph 18: complex space after switching on exciters and PSS Table 9: eigenvalues after switching on the PSS and exciters

Calculated damping and frequencies:

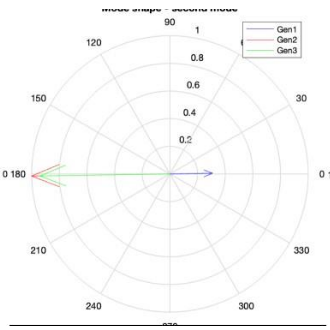
```
damping =

    1.0000
    1.0000
    1.0000
    0.0610
    0.0240
    0.5670
    0.5560
    0.5570
    1.0000
    0.3450
    0.5110
    0.6510
         0
    1.0000
    1.0000
```

```
frequency =

    0.1235
    0.0808
         0
    1.9020
    1.2863
    1.2642
    1.2439
    1.2600
         0
    0.1926
    0.1176
    0.0789
    0.0003
         0
         0
```

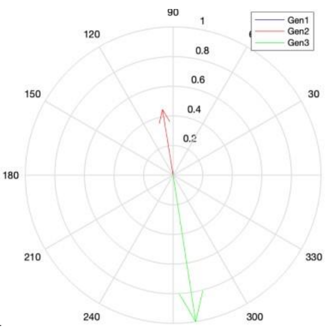
3.2 A) Which generator oscillates against which? Can you classify the modes? Analyze the results by plotting the mode shapes:



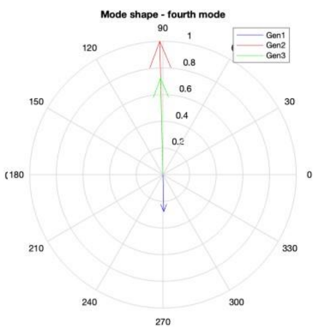
Graph 19: Mode shape-first mode



Graph 20: second mode

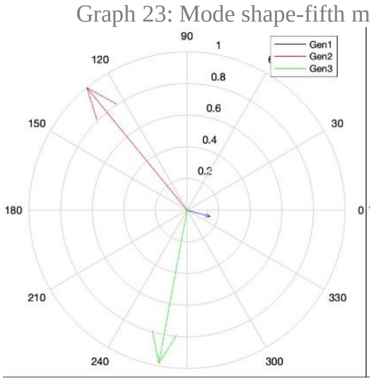


Graph 21: third mode

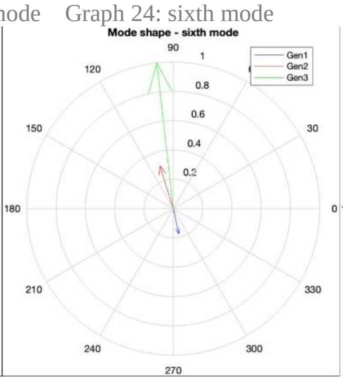


Graph 22: fourth mode

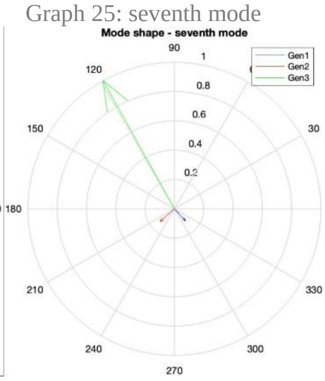
Graph 23: Mode shape-fifth mode



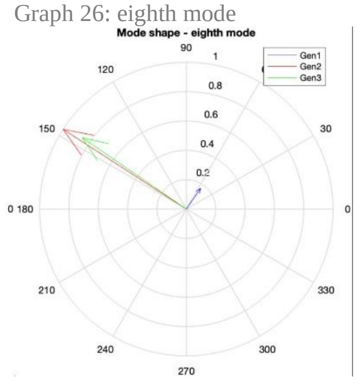
Graph 24: sixth mode

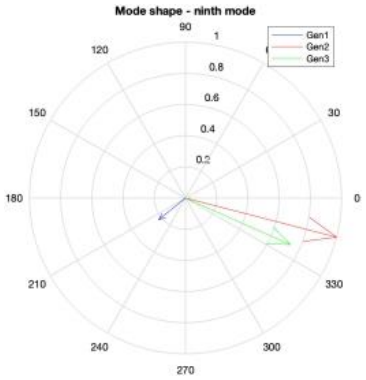


Graph 25: seventh mode

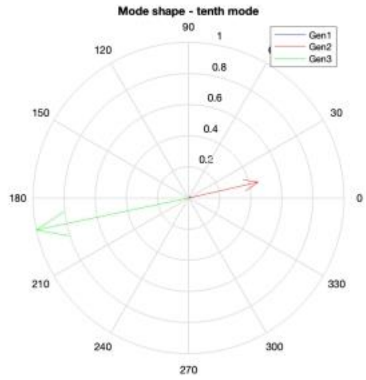


Graph 26: eighth mode





Graph 26: Mode shape-ninth mode



Graph 27: Mode shape-tenth mode

Mode	Dominant States	Oscillation Phase	
1	Δw and Δб of G2, G3	G2	G3
2	Δw and Δб of G1, G2, G3	G2, G3	G1
3	Δw and Δб of G2, G3	G3	G2
4	Δw and Δб of G1, G2, G3	G3, G2	G1
5	Δw and Δб of G1, G2, G3	_	_
6	Δw and Δб of G1, G2, G3	G2	G1
7	Δw and Δб of G1, G2, G3	G3	G1
8	Δw and Δб of G1, G2, G3	_	_
9	Δw and Δб of G1, G2, G3	_	_
10	Δw and Δб of G2, G3	G3	G2
		Anti_Phase	

Table 10

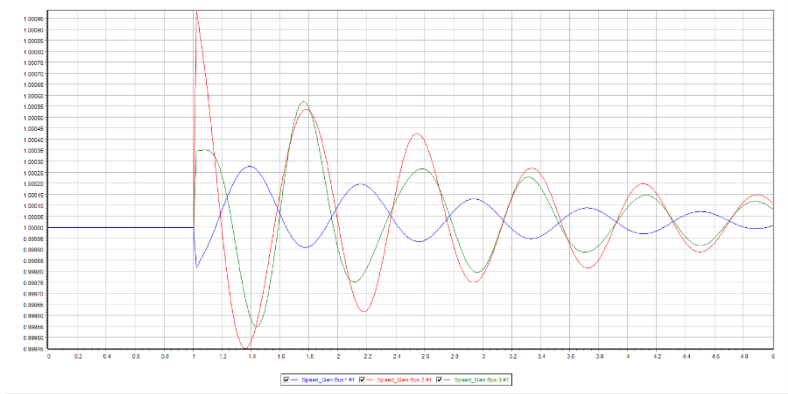
The oscillation modes of the generators are summarized as follows:

- ***Mode 1****:* A local mode, where G3 oscillates out of phase with G2, and G1 has negligible oscillation.
- ***Mode 2****:* An inter-area mode, where G3 and G2 oscillate in phase with each other, and opposite to G1.
- ***Mode 3****:* A local mode, where G3 oscillates out of phase with G2, and G1 is negligible.
- ***Mode 4****:* An inter-area mode, where G2 has a large oscillation, and G1 oscillate in phase with each other and G1 has very small oscillations.
- ***Mode 5****:* A global mode but G1 has very small oscillations.
- ***Mode 6****:* An inter-area mode, where G2 and G1 are out phase of each other and G3 has the highest oscillations.
- ***Mode 7****:* A global mode, but G2 and G1 have small oscillations.
- ***Mode 8****:* An inter-area mode, where G2 and G3 oscillate in phase with each other.
- ***Mode 9:*** An inter-area mode
- ***Mode 10***: A local mode, where G3 oscillates out of phase with G2, and G1 has negligible oscillation.

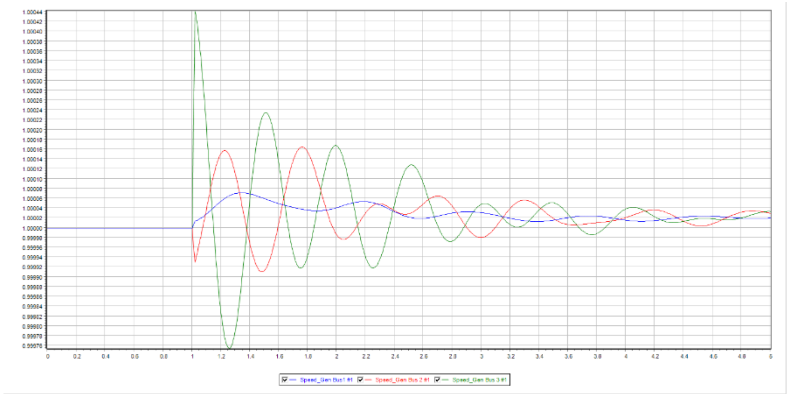
Pkh	Pk1		Pk2				Pk3		Pk4		Pk5		Pk6		Pk7				Pk8		Pk9		Pk10							
G1	4.14E-07	4.14E-07	1.16E-05	1.16E-05		9.64E-09	0.00555	0.00555	0.39143	0.39143	0.00034	0.00034	0.00036	0.00036	0.00023	0.00023	3.08E-06	0.00133	0.00084	0.00084	0.00089	0.00089	8.57E-06	8.57E-06	1	1	0.00175	4.48E-05	1.99E-06	1.47E-15
G2	4.57E-05	4.57E-05	2.64E-05	2.64E-05	2.08E-10	2.09E-10	0.35102	0.35102	1	1	0.00175	0.00175	0.00038	0.00038	0.00021	0.00021	0.01492	0.002	0.00293	0.00293	0.00285	0.00285	0.00157	0.00157	0.19437	0.19437	0.00109	9.16E-06	6.66E-05	4.88E-15
G3	0.00012	0.00012	1.06E-05	1.06E-05	3.24E-10	3.24E-10	1	1	0.29657	0.29657	0.00142	0.00142	0.00088	0.00088	0.00079	0.00079	0.0237	0.01069	0.00212	0.00212	0.00097	0.00097	0.00388	0.00388	0.11246	0.11246	0.00064	7.91E-07	0.00015	8.35E-16
G1	4.14E-07	4.14E-07	1.16E-05	1.16E-05	9.63E-09	9.64E-09	0.00555	0.00555	0.39143	0.39143	0.00034	0.00034	0.00036	0.00036	0.00023	0.00023	3.08E-06	0.00133	0.00084	0.00084	0.00089	0.00089	8.57E-06	8.57E-06	1	1	0.00175	4.48E-05	1.99E-06	3.36E-15
G2	4.57E-05	4.57E-05	2.64E-05	2.64E-05	2.08E-10	2.09E-10	0.35102	0.35102	1	1	0.00175	0.00175	0.00038	0.00038	0.00021	0.00021	0.01492	0.002	0.00293	0.00293	0.00285	0.00285	0.00157	0.00157	0.19437	0.19437	0.00109	9.16E-06	6.66E-05	1.85E-15
G3	0.00012	0.00012	1.06E-05	1.06E-05	3.24E-10	3.24E-10	1	1	0.29657	0.29657	0.00142	0.00142	0.00088	0.00088	0.00079	0.00079	0.0237	0.01069	0.00212	0.00212	0.00097	0.00097	0.00388	0.00388	0.11246	0.11246	0.00064	7.91E-07	0.00015	8.48E-16
G1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
G2	4.68E-06	4.68E-06	5.81E-06	5.81E-06	4.77E-06	4.78E-06	0.01854	0.01854	0.01739	0.01739	0.00432	0.00432	0.00345	0.00345	0.00153	0.00153	1	0.90262	0.04638	0.04638	0.03895	0.03895	0.02804	0.02804	3.27E-09	3.27E-09	2.14E-07	4.37E-07	7.24E-07	1.52E-15
G3	1.33E-05	1.33E-05	5.14E-07	5.14E-07	3.31E-06	3.31E-06	0.09042	0.09042	0.00519	0.00519	0.00152	0.00152	0.00169	0.00169	0.00723	0.00723	0.93585	1	0.02596	0.02596	0.02107	0.02107	0.11343	0.11343	3.36E-09	3.36E-09	1.26E-07	2.99E-07	3.25E-06	1.17E-15
G1	9.77E-06	9.77E-06	0.00012	0.00012	0.00029	0.0003	3.55E-05	3.55E-05	0.0002	0.0002	0.00057	0.00057	0.05099	0.05099	0.00406	0.00406	4.76E-06	0.00301	1	1	1	1	0.00821	0.00821	1.29E-07	1.29E-07	1.77E-05	2.19E-06	1.58E-07	5.80E-19
G2	0.0034	0.0034	0.00509	0.00509	2.01E-06	2.02E-06	0.01979	0.01979	0.04381	0.04381	0.03869	0.03869	0.00679	0.00679	0.0035	0.0035	0.01596	0.00178	0.67022	0.67022	0.77813	0.77813	0.4255	0.4255	3.97E-06	3.97E-06	3.49E-05	1.50E-06	8.51E-06	1.30E-16
G3	0.00781	0.00781	0.00217	0.00217	2.84E-06	2.84E-06	0.03634	0.03634	0.01159	0.01159	0.00139	0.00139	0.01233	0.01233	0.0348	0.0348	0.03017	0.00618	0.38366	0.38366	0.22365	0.22365	1	1	2.10E-06	2.10E-06	1.57E-05	4.25E-07	1.86E-05	4.37E-17
G1	1.00E-05	1.00E-05	0.00012	0.00012	0.0003	0.0003	2.82E-05	2.82E-05	0.00018	0.00018	0.0186	0.0186	0.97327	0.97327	0.20136	0.20136	7.88E-08	7.42E-05	0.15518	0.15518	0.1401	0.1401	0.00112	0.00112	1.11E-08	1.11E-08	6.32E-08	3.50E-08	5.28E-09	2.47E-17
G2	0.00349	0.00349	0.00525	0.00525	2.18E-06	2.18E-06	0.00189	0.00189	0.00148	0.00148	0.98371	0.98371	0.06161	0.06161	0.04581	0.04581	5.58E-05	5.14E-05	0.10163	0.10163	0.1084	0.1084	0.05759	0.05759	1.11E-07	1.11E-07	4.13E-06	4.40E-08	1.91E-07	4.83E-17
G3	0.00799	0.00799	0.00222	0.00222	2.98E-06	2.98E-06	0.00485	0.00485	0.00049	0.00049	0.07286	0.07286	0.17999	0.17999	0.97654	0.97654	6.48E-05	5.50E-05	0.05915	0.05915	0.03233	0.03233	0.14393	0.14393	4.88E-08	4.88E-08	2.12E-06	1.87E-08	4.61E-07	1.47E-17
G1	1.02E-05	1.02E-05	0.00012	0.00012	0.00031	0.00031	2.72E-05	2.72E-05	0.00016	0.00016	0.01921	0.01921	1	1	0.20719	0.20719	7.66E-07	1.82E-06	0.10405	0.10405	0.09148	0.09148	0.00073	0.00073	7.44E-09	7.44E-09	4.21E-08	2.33E-08	3.52E-09	1.14E-18
G2	0.00352	0.00352	0.00531	0.00531	2.20E-06	2.20E-06	0.00184	0.00184	0.0014	0.0014	1	1	0.06241	0.06241	0.04645	0.04645	0.00037	1.74E-05	0.07984	0.07984	0.08403	0.08403	0.04447	0.04447	8.73E-08	8.73E-08	3.24E-06	3.45E-08	1.50E-07	1.10E-17
G3	0.00812	0.00812	0.00226	0.00226	3.02E-06	3.03E-06	0.00469	0.00469	0.00046	0.00046	0.07486	0.07486	0.18406	0.18406	1	1	0.00057	4.53E-06	0.04165	0.04165	0.02227	0.02227	0.09849	0.09849	3.45E-08	3.45E-08	1.48E-06	1.31E-08	3.23E-07	1.35E-18
G1	1.83E-09	1.83E-09	2.18E-08	2.18E-08	5.53E-08	5.54E-08	2.47E-06	2.47E-06	4.23E-05	4.23E-05	0.00548	0.00548	0.30576	0.30576	0.0606	0.0606	1.27E-05	0.02553	0.73997	0.73997	0.77696	0.77696	0.0065	0.0065	4.43E-08	4.43E-08	2.78E-07	1.54E-07	2.32E-08	1.02E-16
G2	6.37E-07	6.37E-07	9.59E-07	9.59E-07	3.98E-10	3.99E-10	0.00017	0.00017	0.00036	0.00036	0.28954	0.28954	0.01936	0.01936	0.01379	0.01379	0.00896	0.01767	0.48461	0.48461	0.60117	0.60117	0.33294	0.33294	4.44E-07	4.44E-07	1.82E-05	1.93E-07	8.41E-07	8.98E-17
G3	1.46E-06	1.46E-06	4.06E-07	4.06E-07	5.44E-10	5.45E-10	0.00043	0.00043	0.00012	0.00012	0.02145	0.02145	0.05655	0.05655	0.2939	0.2939	0.01041	0.01891	0.28204	0.28204	0.17928	0.17928	0.83202	0.83202	1.96E-07	1.96E-07	9.31E-06	8.21E-08	2.03E-06	3.41E-16
G1	1.89E-07	1.89E-07	2.26E-06	2.26E-06	5.71E-06	5.72E-06	1.21E-06	1.21E-06	1.51E-05	1.51E-05	2.52E-05	2.52E-05	0.00011	0.00011	8.82E-05	8.82E-05	3.57E-09	6.05E-07	0.00013	0.00013	0.00024	0.00024	3.57E-06	3.57E-06	6.86E-06	6.86E-06	0.3343	1	0.01385	1.14E-15
G2	6.54E-05	6.54E-05	9.85E-05	9.85E-05	4.08E-08	4.09E-08	0.00028	0.00028	0.0009	0.0009	0.00075	0.00075	2.97E-05	2.97E-05	3.12E-05	3.12E-05	1.57E-05	4.46E-07	0.0003	0.0003	0.00049	0.00049	0.00041	0.00041	1.91E-05	1.91E-05	1	0.20674	0.45715	2.17E-16
G3	0.00015	0.00015	4.19E-05	4.19E-05	5.61E-08	5.62E-08	0.00074	0.00074	0.0003	0.0003	0.00017	0.00017	0.00014	0.00014	0.00093	0.00093	3.54E-05	1.53E-06	0.00015	0.00015	0.00015	0.00015	0.00105	0.00105	1.13E-05	1.13E-05	0.58726	0.01786	1	2.07E-16
G1	0.00125	0.00125	0.0229	0.0229	0.9997	1	7.50E-06	7.50E-06	6.28E-05	6.28E-05	0.00013	0.00013	0.00055	0.00055	0.00045	0.00045	9.89E-09	1.07E-06	8.18E-05	8.18E-05	0.0001	0.0001	1.10E-06	1.10E-06	3.54E-07	3.54E-07	2.63E-05	2.29E-06	1.04E-07	7.59E-16
G2	0.43419	0.43419	0.99996	0.99996	0.00715	0.00715	0.00171	0.00171	0.00375	0.00375	0.00391	0.00391	0.00015	0.00015	0.00016	0.00016	4.36E-05	7.86E-07	0.0002	0.0002	0.00021	0.00021	0.00012	0.00012	9.86E-07	9.86E-07	7.88E-05	4.74E-07	3.44E-06	1.40E-16
G3	0.99991	0.99991	0.42536	0.42536	0.00982	0.00982	0.00458	0.00458	0.00125	0.00125	0.00089	0.00089	0.00072	0.00072	0.00479	0.00479	9.81E-05	2.70E-06	9.97E-05	9.97E-05	6.23E-05	6.23E-05	0.00032	0.00032	5.83E-07	5.83E-07	4.63E-05	4.10E-08	7.53E-06	1.35E-16
G1	0.00125	0.00125	0.0229	0.0229	1	0.9997	9.19E-07	9.19E-07	5.31E-06	5.31E-06	1.16E-05	1.16E-05	4.74E-05	4.74E-05	3.97E-05	3.97E-05	3.22E-10	1.53E-08	1.64E-06	1.64E-06	1.79E-06	1.79E-06	1.85E-08	1.85E-08	7.22E-09	7.22E-09	5.11E-07	4.45E-08	2.02E-09	4.19E-20
G2	0.43423	0.43423	1	1	0.00715	0.00715	0.00021	0.00021	0.00032	0.00032	0.00035	0.00035	1.30E-05	1.30E-05	1.41E-05	1.41E-05	1.42E-06	1.13E-08	3.99E-06	3.99E-06	3.61E-06	3.61E-06	2.10E-06	2.10E-06	2.01E-08	2.01E-08	1.53E-06	9.19E-09	6.67E-08	1.83E-20
G3	1	1	0.42538	0.42538	0.00982	0.00982	0.00056	0.00056	0.00011	0.00011	7.91E-05	7.91E-05	6.20E-05	6.20E-05	0.00042	0.00042	3.19E-06	3.87E-08	2.01E-06	2.01E-06	1.10E-06	1.10E-06	5.43E-06	5.43E-06	1.19E-08	1.19E-08	8.97E-07	7.94E-10	1.46E-07	1.12E-20
	Mode 1		Mode 2				Mode 3		Mode 4		Mode 5		Mode 6		Mode 7				Mode 8		Mode 9		Mode 10							

Table 11: Participation factors after switching exciters and PSS.

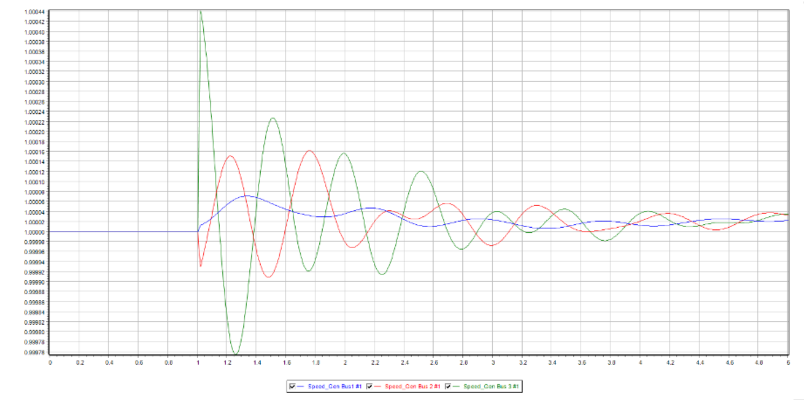
After plotting in the Power World:



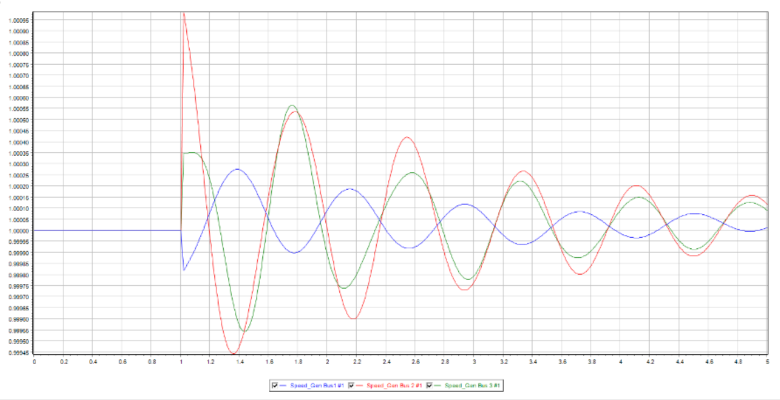
Graph 28: Mode shape-second



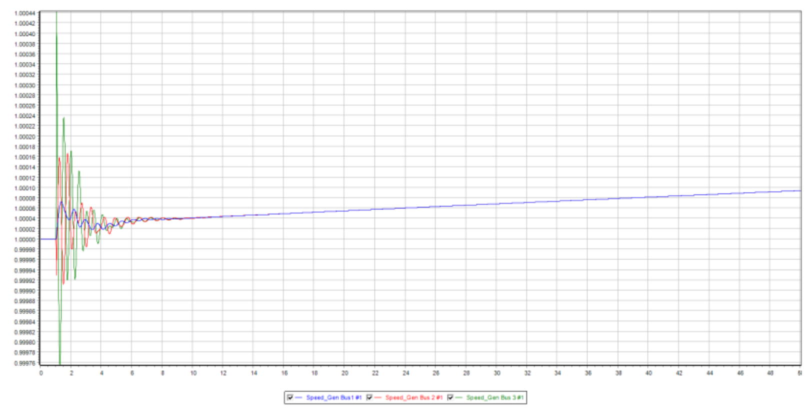
Graph 29: Mode shape-first



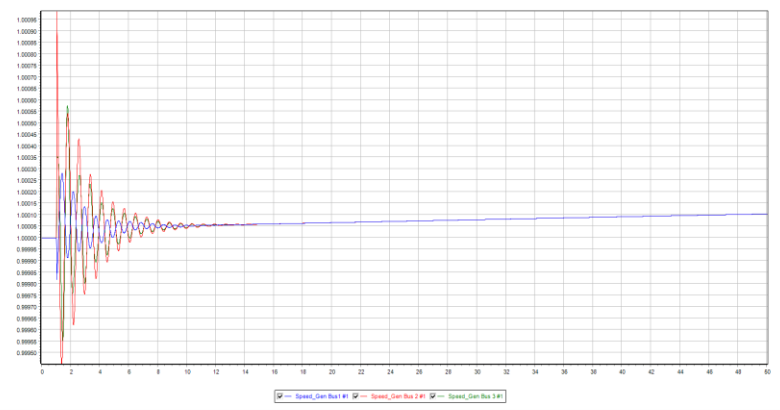
Graph 30: Mode shape-first with ka=50



Graph 31: Mode shape-second with ka=50



Graph 32: 8to9 for 50 sec. after PSS



Graph 33: 5to7 for 50 sec. after PSS

The stability analysis in short and long term for 5-7 and 8-9 shows that both are unstable.

3.3 A) Do you get an improvement in the damping ratios of the electromechanical modes?

The participation factor indicates that the frequency deviation ($\Delta\omega$) and the angle deviation ($\Delta\delta$) are the dominant states for mode 3 and mode 4. The damping ratio for these modes (0.0610 and 0.0240) is unchanged by the application of PSS, which is consistent with case 1 (without AVR and PSS).

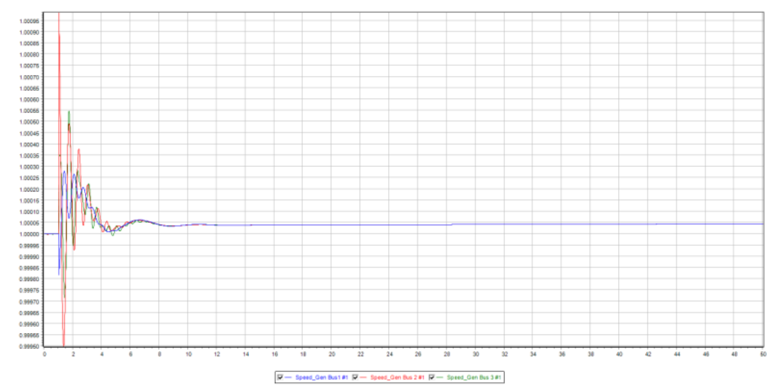
The influence of PSS on the system stability is complex and depends on several factors, such as the location of the generators with PSS, the characteristics and location of the loads, and the exciters of the other units.

3.4 B) Change the gain of the PSS (KS) and try to improve the damping of the electromechanical mode. Comment the results.

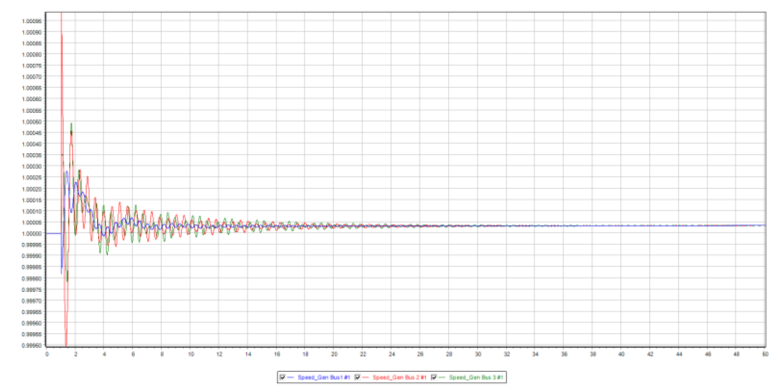
Ks	damping ratio （Ka=20）
0.1(initial)	0.061
1	0.068
1.5	0.071
2	0.074
2.5	0.076
3	0.078
3.5	0.079
4	0.081
4.5	0.081
5	0.082
5.5	0.082
6	0.082
6.5	0.082
7	0.082
7.5	0.081
8	0.081
8.5	0.08
9	0.079
9.5	0.079
10	0.078

Table 11

As Ks increases, the damping ratio initially rises and then falls. The damping ratio peaks at Ks=5 and remains constant until Ks=7. Increasing the PSS gain reduces the recovery time of the system. However, when Ks reaches 20, the recovery time increases again, which matches the MATLAB output.

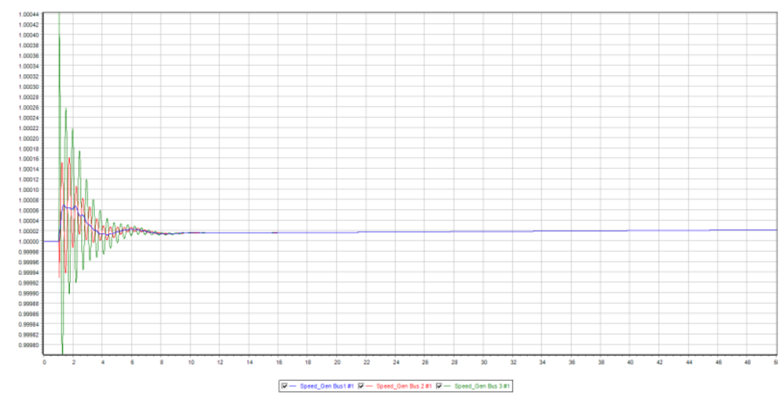


Graph 34: 5 to 7 after Ks=20 for 50 sec

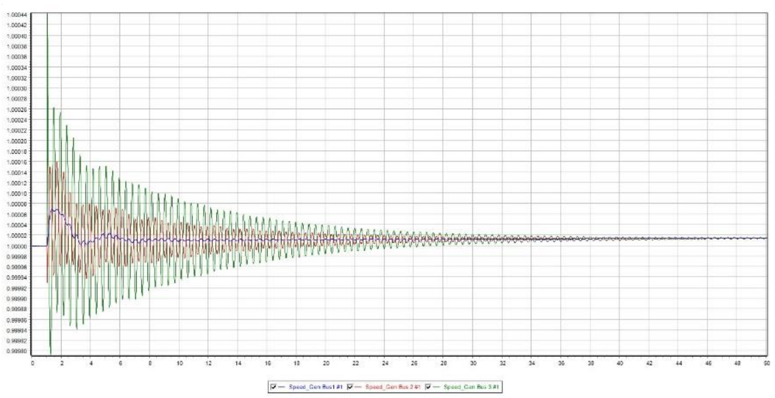


Graph 35: 5to7 after Ks=20 for 50 sec

Graph 36: 8 to 9 after Ks=20 for 50 sec



Graph 37: 8 to 9 after Ks=20 for 50 sec.



As shown in *Table 14*, adjusting the PSS gain (KS) can enhance the damping of the electromechanical mode, which is also consistent with the theoretical analysis:

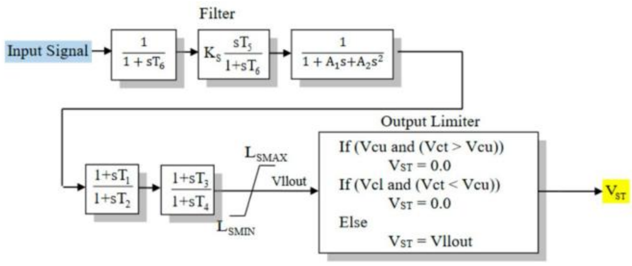


Figure 3. PSS block diagram.

Graph 38

Edited code on WSCC_ANALYSIS.M file:

```
warning off
path(path,'Load_flow');
%                        a                                                AVR   PSS
[A, B, pos_var, ~] = small_sgn_linp('Grid_WSCC_Sauer','gendat_WSCC_9_bus',  1,  0);

[V,eigenvalues] = eig(A); % eigenalaysi
W = adjoint(V)/det(V); % left eigenvectors
% V*W; % test, the result is an unitary matrix: V*W = I

lambda = diag(eigenvalues); % eigenvalues

% taking only one eig./mode
damping = round(-real(lambda(1:2:end))./abs(lambda(1:2:end)),3) % damping
frequency = imag(lambda(1:2:end)/(2*pi)) % frequency
% help: A.' returns the nonconjugate transpose of A
part_fact = abs(V.*W.')./abs(max(V.*W.')); % participation factors (normalized)
%% figures for modal analysis - normalized mode shapes
for i = 1:3 % names
   label = sprintf('Gen%i',i);
   name_gen{i} = label;
end

n = [0 0 1; 1 0 0; 0 1 0]; % color settings

figure
C = compass(V([4,5,6],1)./abs(max(V([4,5,6],1)))); % take the speed index
for k = 1:3 % add the color
        C(k).Color = n(k,:);
end
legend(name_gen);
title('Mode shape - first mode')

%% figures for modal analysis - normalized mode shapes
for i = 1:3 % names
   label = sprintf('Gen%i',i);
   name_gen{i} = label;
end

n = [0 0 1; 1 0 0; 0 1 0]; % color settings

figure
C = compass(V([4,5,6],3)./abs(max(V([4,5,6],3)))); % take the speed index
for k = 1:3 % add the color
        C(k).Color = n(k,:);
end
legend(name_gen);
title('Mode shape - second mode')
% plot
figure
plot(real(round(lambda,3)),imag(lambda),'xw','MarkerSize',12,'Markeredgecolor',[0,0,1],'LineWidth',2.5);
grid on
xlabel('Real'); ylabel('Imag'); title('Complex space');
% legend('DG1=0')
```